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ORIGINAL PAPER

Multiple attribute decision-making based on fuzzy neural network under complex Fermatean fuzzy data

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Abstract
This study proposes a novel fuzzy neural network (FNN) framework operating under complex Fermatean fuzzy sets to enhance multiple attribute decision-making (MADM) in uncertain environments. The model integrates Schweizer–Sklar-based aggregation operators within the FNN's computational layers to process complex fuzzy information, capturing both membership and non-membership degrees. Each layer of the network handles CFF data, enabling adaptive learning and maintaining interpretability. A case study on evaluating innovation in informatization stages demonstrates the model's effectiveness. Results show that the proposed CFNN approach delivers consistent and reliable rankings, reduces subjectivity, and outperforms traditional MADM methods in handling ambiguity and uncertainty.

Keywords Decision making · Schweizer–Sklar t-norm and t-conorm · Aggregation operators · Fuzzy neural network Fermatean fuzzy number · MADM

1 Introduction

These issues are highly relevant in various real-world situations. To reduce investment risk and optimize returns, for instance, an investor might have to choose the best company, or an organization might have to evaluate its degree of information technology before making system upgrades. These choices are classic MADM difficulties, as they typically involve weighing several interconnected elements. Because of its usefulness in day-to-day living and strategic decision-making, MADM has attracted a lot of study interest in recent decades. An intelligent system that combines the reasoning power of fuzzy logic with the learning capabilities of artificial neural networks is called a fuzzy neural network (FNN). Because of this combination, FNNs can accurately and uncertainly describe complicated, nonlinear systems, which are characteristics frequently seen in real-world applications. While the neural network structure allows for automatic tweaking of the fuzzy rules and membership functions through training data, the fuzzy component offers reasoning based on language principles. FNNs are particularly helpful in domains where interpretability and flexibility are crucial, such as control systems, pattern identification, forecasting, and decision-making. They are an effective tool in soft computing because of their capacity to learn from data while retaining human-like thinking. The input signals are expressed as fuzzy numbers, and the transformation from input to output is governed by a set of fuzzy rules.

The structure of the fuzzy neural network is depicted in Fig. 1.

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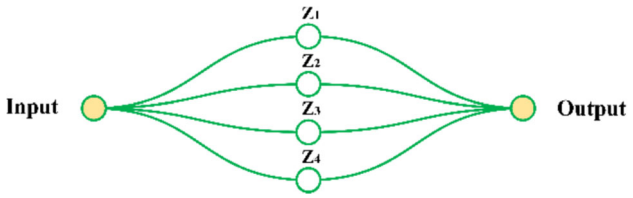


Fig. 1 Structure of fuzzy neural networking

The unconventional interdisciplinary area of tectonophysics uses theories and techniques first created by physicists to solve economic issues, especially those involving nonlinear dynamics, uncertainty, and stochastic processes. Researchers have shown a great deal of interest in these intricate occurrences [1–3]. Fuzzy set theory has become a popular tool among academics to model and resolve complicated financial issues in recent years due to the rising complexity of contemporary economies. Multiple-attribute decision-making is one of its most active application areas. MADM describes situations in which people or groups must assess and select the best option from a range of options based on a number of criteria. These issues are significant in a variety of fields and social settings. To reduce investment risk and optimize returns, for instance, an investor could have to pick the most promising company, and a corporation might have to evaluate its degree of information technology before introducing new technology. Because these choices necessitate a thorough assessment of multiple factors, MADM issues are prevalent and significant in daily life [7, 8]. Consequently, during the past few decades, MADM has gained a lot of attention in practical economics and decision science.

To more efficiently aggregate complicated and interconnected data, Azeem et al. [4] recently suggested complicated Fermatean Fuzzy Partitioned Maclaurin Symmetric Mean (CFFPMSM) operators. Their research presents new aggregation tools that maintain flexibility and symmetry while capturing correlations between input arguments. In the disc spherical fuzzy environment, Ali [5] introduced a WASPAS-based decision assistance method combined with Dombi power aggregation operators. Ali [6] used the probabilistic hesitant fuzzy environment with partitioned Maclaurin symmetric mean operators to offer a unique group decision-making model. In the t-arbicular fuzzy environment, Ali and Popa [9] presented a novel decision-making model that incorporates the TODIM approach with uncertain weights. In the linguistic interval-valued q-Rung orthopair fuzzy environment, Ali et al. [10] presented a novel Grey Relational Analysis (GRA) method based on Choquet integrals.

This approach captures decision-makers' personal risk preferences with regard to innovation as well as their objective evaluations of the specified criteria. Based on the body of existing literature and important aspects of innovation, a rigorous set of criteria is developed to assess the performance of MCDM methods providers in cross-border transactions. We use complex Fermatean fuzzy Schweizer–Sklar operators to deal with uncertainty and inadequate information in the evaluation process. These operators effectively model ambiguity and reluctance resulting from limited knowledge or involvement. The criterion weights are extracted from complex Fermatean fuzzy data using these operators. In order to determine the best choice, the method is also used to calculate an overall performance score for each candidate, placing them in descending order according to the decision-maker's risk tolerance. The standards and techniques for evaluating expertise, stages, crops, and suppliers are succinctly evaluated in this section. This lays the groundwork for developing the selection criteria for major groups or financial institutions that intend to use expertise in the course of cross-border transmission. In order to support complicated decision scenarios comprising several, frequently competing criteria, several Attribute Decision-Making (MADM) procedures are essential. Fuzzy set theory and computer models have been used by scholars in recent years to better represent imprecision and uncertainty in decision-making. Fuzzy neural networks (FNNs) are one of these that have demonstrated promise because of its capacity to learn from and adjust to imprecise input data. Fuzzy neural network-based MADM techniques now in use, however, usually rely on intuitionistic or classical fuzzy sets, which have limitations when it comes to expressing higher levels of uncertainty. Although this expressive capability has been improved by extensions like Pythagorean and q-rung orthopair fuzzy sets, they are still unable to adequately represent more complex decision environments, especially those that involve hesitation and inconsistency. A more comprehensive framework that offers a richer structure to manage complex decision criteria is provided in this situation by Complex Fermatean Fuzzy Sets (CFFS).

The break of this paper is systematized as follows: Sect. 2 remembrances approximately plain notions and provides the SSTT operational rules founded on CFFS. Section 3 write a complex fermatean fuzzy set and operational laws. Section 4 recommends a series of three aggregation operators. Section 5 introduces an original method to MADM and delivers a numerical instance to validate the rationality and qualities of the proposed slant. Section 6 summarizes the paper.

1.1 Literature review

As a result, MAGDM concerns have received a lot of attention over the past few decades [11, 12, 14–17]. Given the complexity of the real creation, fuzzy placed in FS theory, presented by Zadeh in 1965 [13], has received a lot of supplementary and extra attention. Previously, it was calculated to use the membership degree to represent the appearance data

of DMs. By addressing internal skepticism about experts' knowledge, incomplete data, or external illegibility on current risk events, FS theory offers a fresh viewpoint. Following that foundation, Atanassov [1] presented the intuitionistic fuzzy set that adds the basis's no-membership quantity over FS. Because they have been suggested, these twain mystical sets have also been used in dense fields [18–20] and other user MCDM techniques [27]. Fuzzy neural networks (FNNs) [25, 26] were a potent hybrid mathematical model that combines fuzzy logic [24] and artificial neural networks [21–23]. FNNs are well-suited for handling complicated and uncertain data because they combine the learning capabilities of neural networks with fuzzy logic's ability to handle ambiguous or imprecise inputs. The main ideas, characteristics, advantages, and real-world uses of FNNs are described in this section. By combining the advantages of fuzzy logic and neural networks, FNNs offer a more reliable and flexible computational framework. A fuzzy rule foundation, a neural network architecture, input fuzzification, and output defuzzification are the fundamental elements of an FNN. FNNs have drawn a lot of scholarly attention because of their efficacious handling of multi-criteria decision-making (MCDM) problems.

1.2 Research gap

Few studies have integrated fuzzy neural networks with the complicated Fermatean fuzzy environment for MADM problems, despite the growing interest in CFFS. Adaptive and learning-based MADM models in this context are severely lacking, as the majority of recent publications concentrate on standalone aggregation or ranking techniques within the CFF framework.

1.3 Objective of study

This paper suggests a fuzzy neural network-based MADM framework under complex Fermatean fuzzy data in order to close this gap. The suggested approach improves choice dependability in ambiguous contexts by utilizing the improved uncertainty modeling of CFFS in addition to expanding the capabilities of current FNN-based decision models.

2 Preliminaries

Definition 2.1 [13] Let us consider that $\boldsymbol{\Phi} \neq \boldsymbol{X}$ and by a fuzzy set $\boldsymbol{\gamma} = \left\{ \begin{array}{c} \langle \mathbf{x}, \boldsymbol{\mu}_{\boldsymbol{\gamma}(\mathbf{x})} \rangle \\ : \ \mathbf{x} \in \mathbf{X} \end{array} \right\}$, $\boldsymbol{\mu}_{\boldsymbol{\gamma}(\mathbf{x})}$ *i*s a mapping from $\mathbf{X}$ to [0, 1] represent membership function of an element $\mathbf{x}$ in $\mathbf{X}$.

Definition 2.2 [15] The fixed set$C$ and the FFSA is defined in$A = \left\{ \begin{array}{c} \langle \varsigma_A(x), \\ \chi_A(x) \rangle \\ : \ x \in C \end{array} \right\}$, where$\varsigma_A(x)$ and$\chi_A(x)$represent the MED and NOMED, and$\varsigma_A(x) \in [0,1]$, $\chi_A(x) \in [0,1]$and$0 \leq \varsigma_A(x)^3 + \chi_A(x)^3 \leq 1$.The degree of indeterminacy is defined as$\pi_A(x) = \sqrt[3]{(\varsigma_A(x)^3 + \chi_A(x)^3 - \varsigma_A(x)^3 \chi_A(x)^3)}$.The FFS is denoted as $A = \langle \varsigma_A, \chi_A \rangle$.

Definition 2.3 [2] The fixed set$D$and the CFFS$A$is defined in$A = \left\{ \begin{array}{c} \langle \varsigma_A(x) e^{i2\pi(\varsigma_A(x))}, \\ \chi_A(x) e^{i2\pi(\chi_A(x))} \rangle \\ : \ x \in D \end{array} \right\}$, where$\varsigma_A(x) e^{i2\pi(\varsigma_A(x))}and\chi_A(x) e^{i2\pi(\chi_A(x))}$represent the MED and NOMED, and$\varsigma_A(x) e^{i2\pi(\varsigma_A(x))} \in [0,1]$, $\chi_A(x) e^{i2\pi(\chi_A(x))} \in [0,1]and0 \leq \varsigma_A(x)^3 e^{i2\pi(\varsigma_A(x)^3)} + \chi_A(x)^3 e^{i2\pi(\chi_A(x)^3)} \leq 1$.The degree of indeterminacy is defined as

$$\pi_A(x) = \sqrt[3]{(\varsigma_A(x)^3 + \chi_A(x)^3 - \varsigma_A(x)^3 \chi_A(x)^3)}$$

$$e^{i2\pi\left(\sqrt[3]{(\varsigma_A(x)^3 + \chi_A(x)^3 - \varsigma_A(x)^3 \chi_A(x)^3)}\right)}.$$

The CFFS is denoted as $A = \langle \varsigma_A e^{i2\pi(\varsigma_A)}, \chi_A e^{i2\pi(\chi)} \rangle$.

Definition 2.3 [2] Let$a = \left\{ De^{i2\pi(D)}, Ee^{i2\pi(E)} \right\}$be the CFFSs, then the score function is $a = D_\alpha^3 e^{i2\pi(D_\alpha^3)} - E_\alpha^3 e^{i2\pi(E_\alpha^3)}$.

Definition 2.4 [2] Let$a = \left\{ \varsigma e^{i2\pi(\varsigma)}, \chi e^{i2\pi(\chi)} \right\}$be the CFFSs, then the accuracy function is $a = \varsigma_\alpha^3 e^{i2\pi(\varsigma_\alpha^3)} + \chi_\alpha^3 e^{i2\pi(\chi_\alpha^3)}$.

2.1 Schweizer–Sklar t-norm and t-conorm

The definitions of Schweizer–Sklar t-norm and t-conorm are introduced as follows:

$$\begin{aligned} T(g, h) &= (g^\Gamma + h^\Gamma - 1)^{\frac{1}{\Gamma}} T(g, h) \\ &= 1 - ((1-g)^\Gamma + (1-h)^\Gamma - 1)^{\frac{1}{\Gamma}} \end{aligned}$$

where $\Gamma < 0$, $g, h \in [0,1]$. Additionally, when $\Gamma \to 0$, we have $T(g, h) = gh$ and $T(g, h) = g + h - gh$.

They are the algebraic t-norm and algebraic t-conorm.

3 CFFS operational laws founded on Schweizer–Sklar operators

Definition 3.1 Let$L_1 = [s_1 e^{i2\pi(s_1)},\ p_1 e^{i2\pi(p_1)}]andL_2 = [s_2 e^{i2\pi(s_2)},\ p_2 e^{i2\pi(p_2)}]$be two CFFSs based on SST operator and$\Gamma > 0$,

$$L_1 \oplus L_2 = \begin{bmatrix} \sqrt[3]{1-((1-(s_1^3))^{\Gamma}+(1-(s_2^3))^{\Gamma}-1)^{\frac{1}{\Gamma}}} \\ e^{i2\pi\left(\sqrt[3]{1-((1-(s_1^3))^{\Gamma}+(1-(s_2^3))^{\Gamma}-1)^{\frac{1}{\Gamma}}}\right)}, \\ (p_1^{\Gamma}+p_2^{\Gamma}-1)^{\frac{1}{\Gamma}} e^{i2\pi\left((p_1^{\Gamma}+p_2^{\Gamma}-1)^{\frac{1}{\Gamma}}\right)} \end{bmatrix};$$

$$L_1 \otimes L_2 = \begin{bmatrix} (s_1^{\Gamma}+s_2^{\Gamma}-1)^{\frac{1}{\Gamma}} e^{i2\pi\left((s_1^{\Gamma}+s_2^{\Gamma}-1)^{\frac{1}{\Gamma}}\right)}, \\ \sqrt[3]{1-((1-(p_1^3))^{\Gamma}+(1-(p_2^3))^{\Gamma}-1)^{\frac{1}{\Gamma}}} \\ e^{i2\pi\left(\sqrt[3]{1-((1-(p_1^3))^{\Gamma}+(1-(p_2^3))^{\Gamma}-1)^{\frac{1}{\Gamma}}}\right)} \end{bmatrix};$$

$$nL_1 = \begin{bmatrix} \sqrt[3]{1-(n(1-(s_1^3))^{\Gamma}+(n-1))^{\frac{1}{\Gamma}}} \\ e^{i2\pi\left(\sqrt[3]{1-(n(1-(s_1^3))^{\Gamma}+(n-1))^{\frac{1}{\Gamma}}}\right)}, \\ (np_1^{\Gamma}+(n-1))^{\frac{1}{\Gamma}} e^{i2\pi\left((np_1^{\Gamma}+(n-1))^{\frac{1}{\Gamma}}\right)} \end{bmatrix};$$

$$a_1^n = \begin{bmatrix} (ns_1^{\Gamma}+(n-1))^{\frac{1}{\Gamma}} e^{i2\pi\left((ns_1^{\Gamma}+(n-1))^{\frac{1}{\Gamma}}\right)}, \\ \sqrt[3]{1-(n(1-(p_1^3))^{\Gamma}+(n-1))^{\frac{1}{\Gamma}}} \\ e^{i2\pi\left(\sqrt[3]{1-(n(1-(p_1^3))^{\Gamma}+(n-1))^{\frac{1}{\Gamma}}}\right)} \end{bmatrix}.$$

4 CFF data based on aggregation operators

In this section, CFFSSTWA, CFFSSTOWA, and CFFSSTHWA operators are defined.

4.1 CFFSSTWA operator

Definition 4.1.1 The collection of CFFSs are$d_L = [s_L e^{i2\pi(s_L)},\ p_L e^{i2\pi(p_L)}]$and the weight vector is$f = (f_1, f_2, ..., f_L)^T$with$f_L \in [0,1]$and $\sum_{L=1}^{n} f_L = 1$. Then CFFSST $(d_1, d_2, ..., d_L) = \oplus_{L=1}^{n} f_L d_L$ is said CFFSSTWA operator.

Theorem 4.1.2 The collection of CFFSs are $d_j = [Le^{i2\pi(L_j)},\ Me^{i2\pi(M_L)}]$and the weight vector is $\vartheta = (\vartheta_1, \vartheta_2, ..., \vartheta_n)^T$with $\vartheta_j \in [0,1]$and $\sum_{j=1}^{n} \vartheta_j = 1$. Then it is said CFFSSTWA operator and

$$CFFSSTWA\,(d_1, d_2, ..., d_n) = \begin{bmatrix} \sqrt[3]{1-(\rho(1-(L_1^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}} \\ e^{i2\pi\left(\sqrt[3]{1-\prod^{j=1}_{n}(\rho(1-(L_1^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\right)}, \\ (\rho M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}} e^{i2\pi\left((\rho M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}\right)} \end{bmatrix}.$$

Proof Since n is true and $n = 1$

$$\vartheta_1 d_1 = \begin{bmatrix} \sqrt[3]{1-\prod_{j=1}^{1}(\rho(1-(L_1^3))^{\vartheta_1}+(\rho-1))^{\frac{1}{\vartheta_1}}} \\ e^{i2\pi\left(\sqrt[3]{1-\prod_{j=1}^{1}(\rho(1-(L_1^3))^{\vartheta_1}+(\rho-1))^{\frac{1}{\vartheta_1}}}\right)}, \\ (\rho\prod_{j=1}^{1} M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta_1}} \\ e^{i2\pi\left((\rho\prod_{j=1}^{1} M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta_1}}\right)} \end{bmatrix}$$

$$\vartheta_2 d_2 = \begin{bmatrix} \sqrt[3]{1-\prod_{j=1}^{2}(\rho(1-(L_2^3))^{\vartheta_2}+(\rho-1))^{\frac{1}{\vartheta_1}}} \\ e^{i2\pi\left(\sqrt[3]{1-\prod_{j=1}^{2}(\rho(1-(L_2^3))^{\vartheta_2}+(\rho-1))^{\frac{1}{\vartheta_1}}}\right)}, \\ (\rho\prod_{j=1}^{2} M_2^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta_2}} \\ e^{i2\pi\left((\rho\prod_{j=1}^{2} M_2^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta_2}}\right)} \end{bmatrix}$$

$$\vartheta_1 d_1 \oplus \vartheta_2 d_2 = \begin{bmatrix} \sqrt[3]{1-\prod_{j=1}^{1}(\rho(1-(L_1^3))^{\vartheta_1}+(\rho-1))^{\frac{1}{\vartheta_1}}} \\ e^{i2\pi\left(\sqrt[3]{1-\prod_{j=1}^{1}(\rho(1-(L_1^3))^{\vartheta_1}+(\rho-1))^{\frac{1}{\vartheta_1}}}\right)}, \\ (\rho\prod_{j=1}^{1} M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta_1}} \\ e^{i2\pi\left((\rho\prod_{j=1}^{1} M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta_1}}\right)} \end{bmatrix} \oplus$$

$$\left[\begin{array}{c}\sqrt[3]{1-\prod_{j=1}^{2}(\rho(1-(L_2^3))^{\vartheta_2}+(\rho-1))^{\frac{1}{\vartheta_1}}}\\ e^{i2\pi\left(\sqrt[3]{1-\prod_{j=1}^{2}(\rho(1-(L_2^3))^{\vartheta_2}+(\rho-1))^{\frac{1}{\vartheta_1}}}\right)},\\ (\rho\prod_{j=1}^{2}M_2^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta_2}}\\ e^{i2\pi\left((\rho\prod_{j=1}^{2}M_2^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta_2}}\right)}\end{array}\right]$$

Since $n = k$.
CFFSSTWA $(a_1,\ a_2,\ ...,\ a_n) =$

$$\left[\begin{array}{c}\sqrt[3]{1-\prod_{j=1}^{k}(\rho(1-(L_1^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\\ e^{i2\pi\left(\sqrt[3]{1-\prod_{j=1}^{k}(\rho(1-(L_1^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\right)},\\ (\rho\prod_{j=1}^{k}M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}\\ e^{i2\pi\left((\rho\prod_{j=1}^{k}M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}\right)}\end{array}\right]$$

Since $n = k + 1$.
CFFSSTWA $(a_1,\ a_2,\ ...,\ a_n) =$

$$\left[\begin{array}{c}\sqrt[3]{1-\prod_{j=1}^{k+1}(\rho(1-(L_1^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\\ e^{i2\pi\left(\sqrt[3]{1-\prod_{j=1}^{k+1}(\rho(1-(L_1^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\right)},\\ (\rho\prod_{j=1}^{k+1}M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}\\ e^{i2\pi\left((\rho\prod_{j=1}^{k+1}M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}\right)}\end{array}\right]$$

Theorem 4.1.3 *(Idempotency)* If $\widehat{Z} = [de^{i2\pi(d)},\ \varsigma e^{i2\pi(\varsigma)}]$for all $j = 1,\ 2,\ 3,\ ...,\ n$, then $CFFSSTWA(Z,\ Z,\ Z,\ ...,\ Z) = Z$.

Proof Since $\widehat{Z} = [de^{i2\pi(d)},\ \varsigma e^{i2\pi(\varsigma)}]$

$$= \left[\begin{array}{c}\sqrt[3]{1-\prod_{j=1}^{n}(\rho(1-(d_j^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\\ e^{i2\pi\left(\sqrt[3]{1-\prod_{j=1}^{n}(\rho(1-(d_j^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\right)},\\ (\rho\prod_{j=1}^{n}\varsigma_j^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}e^{i2\pi\left((\rho\prod_{j=1}^{n}\varsigma_j^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}\right)}\end{array}\right]$$

$$= \left[\begin{array}{c}\sqrt[3]{1-(\rho(1-(d_j^3))^{\sum_{j=1}^{n}\vartheta_j}+(\rho-1))^{\frac{1}{\vartheta}}}\\ e^{i2\pi\left(\sqrt[3]{1-(\rho(1-(d_j^3))^{\sum_{j=1}^{n}\vartheta_j}+(\rho-1))^{\frac{1}{\vartheta}}}\right)},\\ (\rho\varsigma_j^{\sum_{j=1}^{n}\vartheta_j}+(\rho-1))^{\frac{1}{\vartheta}}e^{i2\pi\left((\rho\varsigma_j^{\sum_{j=1}^{n}\vartheta_j}+(\rho-1))^{\frac{1}{\vartheta}}\right)}\end{array}\right]$$

$$= \left[\begin{array}{c}\sqrt[3]{1-(\rho(1-(d_j^3))+(\rho-1))^{\frac{1}{\vartheta}}}\\ e^{i2\pi\left(\sqrt[3]{1-(\rho(1-(d_j^3))+(\rho-1))^{\frac{1}{\vartheta}}}\right)},\\ (\rho\varsigma_j+(\rho-1))^{\frac{1}{\vartheta}}e^{i2\pi\left((\rho\varsigma_j+(\rho-1))^{\frac{1}{\vartheta}}\right)}\end{array}\right]$$

CFFSSTWA $(Z,\ Z,\ Z,\ ...,\ Z) = Z$.

4.2 CFFSSTOWA operator

Definition 4.2.1 The collection of CFFSs are $d_L = [s_L e^{i2\pi(s_L)},\ p_L e^{i2\pi(p_L)}]$ and the weight vector is $f = (f_1,\ f_2,\ ...,\ f_L)^T$ with $f_L \in [0,1]$ and $\sum_{L=1}^{n} f_L = 1$. Then CFFSST $(d_1,\ d_2,\ ...,\ d_L) = \oplus_{L=1}^{n} f_L d_L$ is said CFFSSTWA operator.

Theorem 4.2.2 The collection of CFFSs are $d_j = [Le^{i2\pi(L_j)},\ Me^{i2\pi(M_L)}]$and the weight vector is $\vartheta = (\vartheta_1,\ \vartheta_2,\ ...,\ \vartheta_n)^T$ with $\vartheta_j \in [0,1]$and $\sum_{j=1}^{n}\vartheta_j = 1$. Then it is said CFFSSTWA operator and $CFFSSTWA(d_1,\ d_2,\ ...,\ d_n)$

$$= \left[\begin{array}{c}\sqrt[3]{1-(\rho(1-(L_1^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\\ e^{i2\pi\left(\sqrt[3]{1-(\rho(1-(L_1^3))^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}}\right)},\\ (\rho M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}e^{i2\pi\left((\rho M_1^{\vartheta}+(\rho-1))^{\frac{1}{\vartheta}}\right)}\end{array}\right].$$

Proof This proof is the same as in Theorem 4.1.2.

Theorem 4.2.3 *(Idempotency)* If $\widehat{Z} = [de^{i2\pi(d)},\ \varsigma e^{i2\pi(\varsigma)}]$for all $L = 1,2,\ 3,\ ...,\ m$, then $CFFSSTOWA(Z,\ Z,\ Z,\ ...,\ Z) = Z$.

Proof This proof is the same as in Theorem 4.1.3

4.3 CFFSSTHWA operator

Definition 4.3.1 The collection of CFFSs are$d_L = [s_L e^{i2\pi(s_L)},\ p_L e^{i2\pi(p_L)}]$and the weight vector is $f = (f_1,\ f_2,\ ...,\ f_L)^T$ with $f_L \in [0,1]$and $\sum_{L=1}^{n} f_L = 1$. Then CFFSST

$(d_1, d_2, ..., d_L) = \oplus_{L=1}^{n} f_L d_L$ is said CFFSSTHWA operator.

Theorem 4.3.2 The collection of CFFSs are $d_j = [Le^{i2\pi(L_j)}, Me^{i2\pi(M_L)}]$and the weight vector is $\vartheta = (\vartheta_1, \vartheta_2, ..., \vartheta_n)^T$with $\vartheta_j \in [0,1]$and $\sum_{j=1}^{n} \vartheta_j = 1$. Then it is said the CFFSSTHWA operator and

$$CFFSSTHWA(d_1, d_2, ..., d_n) = \begin{bmatrix} \sqrt[3]{1 - \prod_{j=1}^{n} (\rho(1 - (L_1^3))^{\vartheta} + (\rho - 1))^{\frac{1}{\vartheta}}} \\ e^{i2\pi\left(\sqrt[3]{1 - \prod_{j=1}^{n} (\rho(1-(L_1^3))^{\vartheta} + (\rho-1))^{\frac{1}{\vartheta}}}\right)}, \\ (\rho \prod_{j=1}^{n} M_1^{\vartheta} + (\rho - 1))^{\frac{1}{\vartheta}} \\ e^{i2\pi\left((\rho \prod_{j=1}^{n} M_1^{\vartheta} + (\rho-1))^{\frac{1}{\vartheta}}\right)} \end{bmatrix}.$$

Proof This proof is the same as in Theorem 4.1.2.

Theorem 4.3.3 *(Idempotency)* If $\widehat{Z} = [de^{i2\pi(d)}, \varsigma e^{i2\pi(\varsigma)}]$for all $L = 1,2, 3, ..., m$, then $CFFSSTHWA(Z, Z, Z, ..., Z) = Z$.

Proof: This proof is the same as in Theorem 4.1.3.

5 CFF-AHP for innovativeness procedure selection

In order to address the challenging task of choosing innovation service providers for major financial institutions, this section presents a novel integrated approach that combines the Analytic Hierarchy Process with the Schweizer–Sklar Triangular operator in a Complex Fermatean Fuzzy environment. In order to capture both performance strengths and possible dangers, a hierarchical structure of evaluation criteria is first designed based on a thorough literature review and the unique characteristics of innovative service providers. We have updated the manuscript to clarify that Schweizer–Sklar operators give a parametric model, under which the behavior of aggregation can be controlled between widely utilized norms like the minimum and product. Having such flexibility makes it possible to more precisely capture the interrelations among input values, particularly in settings with varying amounts of uncertainty and hesitancy, like those represented by Complex Fermatean Fuzzy Sets. We also drew from literature that illustrates the higher discriminative capability and flexibility of Schweizer–Sklar operators over fixed norms. Such attributes give rise to higher ranking consistencies and decision precision in Multiple Attribute Decision-Making (MADM) problems.

In this paper, we chose four criteria and four alternatives to represent a realistic and feasible decision-making problem applicable to the selected application domain (e.g., evaluation of service providers, technology choice, etc.). The choice was made based on the following criteria:

The four criteria were selected following a review of the pertinent literature and consultation with subject experts. The criteria are each key to the decision problem and together provide complete coverage of all issues without duplication.

Restricting the number of criteria and options enables us to concentrate on proving the efficiency and interpretability of the suggested CFF fuzzy neural network model without confusing the reader with an overload of data or computational detail.

Since this research presents a new hybrid decision-making approach, a simple example with four options and criteria was considered adequate to demonstrate the method. This length is adequate to balance practical applicability and computational clarity.

Although we employed a limited example to illustrate the steps, the approach is completely scalable to larger decision matrices. We have taken care to keep the model robust and effective for larger real-world problems with more criteria and alternatives.

Innovativeness selection, as a characteristic MCDM problem, is applied through a collection of decision matrices. Let $U = \{U_1, U_2, ..., U_n\}$ represent sure latent skill providers $LIS = \{LIS_1, LIS_2, ..., LIS_n\}$ denote significant evaluative criteria delivered by numerous experts $DM = \{DM_1, DM_2, ..., DM_n\}$, and the corresponding rank of specialists is denoted $\Gamma = (\Gamma_1, \Gamma_2, ..., \Gamma_n)^T$ satisfying the environments $\sum_{j=1}^{n} \Gamma_j = 1$. The decision-maker's assessment of a particular applicant on a particular criterion is represented by each entry in the decision matrix.

Input Layer

Step 1: Explain the structure of the CFF decision matrix.

$$\text{IP} = \begin{bmatrix} \begin{bmatrix} s_{11}e^{i2\pi(s_{11})}, \\ p_{11}e^{i2\pi(p_{11})} \end{bmatrix} \begin{bmatrix} s_{12}e^{i2\pi(s_{12})}, \\ p_{12}e^{i2\pi(p_{12})} \end{bmatrix} \cdots\cdots \begin{bmatrix} s_{1n}e^{i2\pi(s1n)}, \\ p_{1n}e^{i2\pi(p_{1n})} \end{bmatrix} \\ \begin{bmatrix} s_{21}e^{i2\pi(s_L)}, \\ p_{21}e^{i2\pi(p_L)} \end{bmatrix} \begin{bmatrix} s_{22}e^{i2\pi(s_L)}, \\ p_{22}e^{i2\pi(p_L)} \end{bmatrix} \cdots\cdots \begin{bmatrix} s_{2n}e^{i2\pi(s_{2n})}, \\ p_{2n}e^{i2\pi(p_{2n})} \end{bmatrix} \\ \begin{bmatrix} s_{m1}e^{i2\pi(s_{m1})}, \\ p_{m1}e^{i2\pi(p_{m1})} \end{bmatrix} \begin{bmatrix} s_{m2}e^{i2\pi(s_{m2})}, \\ p_{m2}e^{i2\pi(p_{m2})} \end{bmatrix} \cdots\cdots \begin{bmatrix} s_{2m}e^{i2\pi(s_{2m})}, \\ p_{2m}e^{i2\pi(p_{2m})} \end{bmatrix} \end{bmatrix}$$

Hidden layer

Step 1: To calculate the hidden layer information of the CFF neural network, we use the CFF SSTHWA operator to aggregate the input signals with the related weight.

Step 2: We use the following formulas to determine the hidden layer criteria weights such that $\sum_{j=1}^{n} \vartheta_j = 1$

Output layer

Step 1: To calculate the hidden layer information of the CFF neural network, we use the CFF SSTHWA operator to aggregate the input signals with the related weights.

Step 2: We combine the hidden layer information with the associated weights using the CFFSSTHWA operator to calculate the output layer information of the CF fuzzy neural network.

Step 3: The output layer generates score values by applying a score function.

Step 4: Determine the ranking order.

Pseudocode of the Proposed CFFNN-Based MADM Algorithm

Inputs

m alternatives:$A = \{A_1, A_2, \ldots . A_m\}$

n criteria $C = \{C_1, C_2, \ldots . C_n\}$

CFF decision matrix: IP=$[CFF_{ij}]$ where

$CFF_{ij} = \left(S_{ij}e^{i2\pi_{ij}}, p_{ij}e^{i2\pi_{ij}}\right)$ for $i = 1\ to\ m, j = 1\ to\ n$

Criteria weight vector: $W = \{w_1, w_2, \ldots . w_n\}$

Output:

Score values and ranking of alternatives

Begin

Step 1: Construct the CFF Decision Matrix

For each alternative A_i $(i = 1\ to\ m)$ each criterion C_j $(j = 1\ to\ n)$

$$CFF_{ij} = \left(S_{ij}e^{i2\pi_{ij}}, p_{ij}e^{i2\pi_{ij}}\right)$$

Hidden layer

Step 1: Hidden Layer Aggregation using CFF-SSTHWA

For each alternative A_i

Aggregate CFF inputs with criteria weights:

$$H_i = CFFSSTHWA(CFF_{i1}, CFF_{i2}, \ldots . CFF_{in})$$

Step 2: Normalize Hidden Layer Weights

Ensure the hidden layer weights $\sum_{j=1}^{n} \vartheta_j = 1$

Output layer

Step 1 and 2

Output Layer Aggregation using CFF-SSTHWA

For each H_i

Compute final output

$$\theta_i = CFFSSTHWA(H_i, \theta_1, \theta_2, \ldots \theta_n)$$

Step 3:Score Function Application

For each alternative A_i

Compute score value using the CFF score function: Score$Score_i = S(0_i)$

Step 4: Ranking of Alternatives

Rank all alternatives A_i to A_m in descending order based on Score$_i$

End

Pseudocode of the Proposed CFFNN-Based MADM Algorithm

5.1 CFF-AHP case history

The objective of this case study is to assess three possible renewable energy projects solar, wind, and biomass under ambiguous, imprecise, and subjective circumstances by a government energy committee. Initial investment cost, environmental effect, energy efficiency, scalability, and societal acceptance are the four main evaluation factors. In order to model not only the degrees of membership and non-membership but also the reluctance and uncertainty using complicated numbers, expert opinions from engineers, economists, and environmental scientists are collected in the form of complex Fuzzy data. Compared to conventional fuzzy systems, this improved modeling capability more precisely captures the ambiguity and inconsistent nature of decision-makers' assessments. A unique fuzzy neural network (FNN) is constructed and trained to process the gathered data and score the projects. The network structure consists of an output layer that provides the final judgment scores, a hidden layer that performs fuzzy aggregation and learning, and an input layer that represents characteristics. The FNN's learning rules and activation functions are modified to function by sophisticated Fermatean fuzzy number arithmetic rules. Even in cases when the input data contains significant amounts of ambiguity or inconsistency, the application of this sophisticated fuzzy neural network enables the fusion of adaptive learning with human-like reasoning, producing more dependable and interpretable results. The case study shows how intelligent neural computing and complex Fuzzy-Neural modeling greatly improve the quality of decision-making in multi-criteria settings, particularly in real-world situations with contradicting and unclear evidence. To improve urban mobility, the planning department of a large city is assessing four options for smart transportation systems: (HIC_1) AI-Based Ride-Sharing Networks, (HIC_2) Electric Trams, (HIC_3) Hyperloop Pods, and (HIC_4) Autonomous Buses. Particularly in areas involving future technologies, the review must reflect the inherent uncertainties in expert opinions and citizen feedback. This is addressed by gathering expert evaluations in the form of complicated Fuzzy data, which offer a strong framework for representing degrees of membership, non-membership, and reluctance as well as the intricate uncertainty of each value. Four main criteria form the basis of the decision:

LIS_1: Cost Efficiency: The sum of setup and running expenses. LIS_2: Sustainability of the Environment: Effects on emissions, pollution, and ecological footprint. LIS_3: Implementation Feasibility: The ability to deploy within the current municipal infrastructure with ease, both legally and technically. LIS_4: Public Acceptance: Surveys are used to estimate social adaptation and user happiness operator to aggregate the input signals with the related weights.

Figure 2 illustrates the workflow of the proposed fuzzy

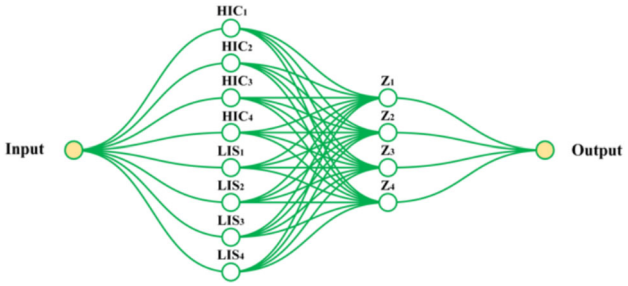


Fig. 2 Schematic representation of the decision-making model

neural network-based decision-making model.

Input Layer

Step 1: Define the CFF decision matrix Table 1.

CFF decision matrix in Table 1

	HIC_1	HIC_2
LIS_1	$\begin{bmatrix} 0.1e^{i2\pi(0.1)}, \\ 0.3e^{i2\pi(0.3)} \end{bmatrix}$	$\begin{bmatrix} 0.05e^{i2\pi(0.05)}, \\ 0.7e^{i2\pi(0.7)} \end{bmatrix}$
LIS_2	$\begin{bmatrix} 0.05e^{i2\pi(0.05)}, \\ 0.7e^{i2\pi(0.7)} \end{bmatrix}$	$\begin{bmatrix} 0.1e^{i2\pi(0.1)}, \\ 0.3e^{i2\pi(0.3)} \end{bmatrix}$
LIS_3	$\begin{bmatrix} 0.04e^{i2\pi(0.04)}, \\ 0.8e^{i2\pi(0.8)} \end{bmatrix}$	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.04e^{i2\pi(0.04)} \end{bmatrix}$
LIS_4	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.04e^{i2\pi(0.04)} \end{bmatrix}$	$\begin{bmatrix} 0.04e^{i2\pi(0.04)}, \\ 0.8e^{i2\pi(0.8)} \end{bmatrix}$

	HIC_3	HIC_4
LIS_1	$\begin{bmatrix} 0.04e^{i2\pi(0.04)}, \\ 0.8e^{i2\pi(0.8)} \end{bmatrix}$	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.04e^{i2\pi(0.04)} \end{bmatrix}$
LIS_2	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.04e^{i2\pi(0.04)} \end{bmatrix}$	$\begin{bmatrix} 0.04e^{i2\pi(0.04)}, \\ 0.8e^{i2\pi(0.8)} \end{bmatrix}$
LIS_3	$\begin{bmatrix} 0.05e^{i2\pi(0.05)}, \\ 0.7e^{i2\pi(0.7)} \end{bmatrix}$	$\begin{bmatrix} 0.1e^{i2\pi(0.1)}, \\ 0.3e^{i2\pi(0.3)} \end{bmatrix}$
LIS_4	$\begin{bmatrix} 0.1e^{i2\pi(0.1)}, \\ 0.3e^{i2\pi(0.3)} \end{bmatrix}$	$\begin{bmatrix} 0.05e^{i2\pi(0.05)}, \\ 0.7e^{i2\pi(0.7)} \end{bmatrix}$

CFF decision matrix in Table 2.

	HIC_1	HIC_2
LIS_1	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.03e^{i2\pi(0.03)} \end{bmatrix}$	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.03e^{i2\pi(0.03)} \end{bmatrix}$
LIS_2	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.03e^{i2\pi(0.03)} \end{bmatrix}$	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.03e^{i2\pi(0.03)} \end{bmatrix}$
LIS_3	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.03e^{i2\pi(0.03)} \end{bmatrix}$	$\begin{bmatrix} 0.3e^{i2\pi(0.3)}, \\ 0.5e^{i2\pi(0.5)} \end{bmatrix}$
LIS_4	$\begin{bmatrix} 0.3e^{i2\pi(0.3)}, \\ 0.5e^{i2\pi(0.5)} \end{bmatrix}$	$\begin{bmatrix} 0.04e^{i2\pi(0.04)}, \\ 0.8e^{i2\pi(0.8)} \end{bmatrix}$

	HIC_3	HIC_4
LIS_1	$\begin{bmatrix} 0.024e^{i2\pi(0.024)}, \\ 0.028e^{i2\pi(0.028)} \end{bmatrix}$	$\begin{bmatrix} 0.3e^{i2\pi(0.3)}, \\ 0.5e^{i2\pi(0.5)} \end{bmatrix}$
LIS_2	$\begin{bmatrix} 0.3e^{i2\pi(0.3)}, \\ 0.5e^{i2\pi(0.5)} \end{bmatrix}$	$\begin{bmatrix} 0.0024e^{i2\pi(0.0024)}, \\ 0.0028e^{i2\pi(0.0028)} \end{bmatrix}$
LIS_3	$\begin{bmatrix} 0.05e^{i2\pi(0.05)}, \\ 0.7e^{i2\pi(0.7)} \end{bmatrix}$	$\begin{bmatrix} 0.1e^{i2\pi(0.1)}, \\ 0.3e^{i2\pi(0.3)} \end{bmatrix}$
LIS_4	$\begin{bmatrix} 0.2e^{i2\pi(0.2)}, \\ 0.3e^{i2\pi(0.3)} \end{bmatrix}$	$\begin{bmatrix} 0.02e^{i2\pi(0.02)}, \\ 0.7e^{i2\pi(0.7)} \end{bmatrix}$

Hidden layer

Step 1: To calculate the hidden layer information of the CFF neural network, we use the CFF SSTHWA

Table 3, CFF SSTHWA operator

	HIC_1	HIC_2
LIS_1	$\begin{bmatrix} 0.1091e^{i2\pi(0.0245)}, \\ 0.3881e^{i2\pi(0.0212)} \end{bmatrix}$	$\begin{bmatrix} 0.3092e^{i2\pi(0.0456)}, \\ 0.5066e^{i2\pi(0.0269)} \end{bmatrix}$
LIS_2	$\begin{bmatrix} 0.1523e^{i2\pi(0.1698)}, \\ 0.1995e^{i2\pi(0.0298)} \end{bmatrix}$	$\begin{bmatrix} 0.0031e^{i2\pi(0.0172)}, \\ 0.8162e^{i2\pi(0.0662)} \end{bmatrix}$
LIS_3	$\begin{bmatrix} 0.1916e^{i2\pi(0.0217)}, \\ 0.2361e^{i2\pi(0.0126)} \end{bmatrix}$	$\begin{bmatrix} 0.6891e^{i2\pi(0.0963)}, \\ 0.7784e^{i2\pi(0.1297)} \end{bmatrix}$
LIS_4	$\begin{bmatrix} 0.0341e^{i2\pi(0.1234)}, \\ 0.0675e^{i2\pi(0.5431)} \end{bmatrix}$	$\begin{bmatrix} 0.3696e^{i2\pi(0.4596)}, \\ 0.5968e^{i2\pi(0.0369)} \end{bmatrix}$

	HIC_3	HIC_4
LIS_1	$\begin{bmatrix} 0.0198e^{i2\pi(0.0198)}, \\ 0.4911e^{i2\pi(0.0563)} \end{bmatrix}$	$\begin{bmatrix} 0.2991e^{i2\pi(0.0297)}, \\ 0.9913e^{i2\pi(0.1298)} \end{bmatrix}$
LIS_2	$\begin{bmatrix} 0.8031e^{i2\pi(0.0278)}, \\ 0.9142e^{i2\pi(0.0212)} \end{bmatrix}$	$\begin{bmatrix} 0.7402e^{i2\pi(0.7402)}, \\ 0.9741e^{i2\pi(0.9741)} \end{bmatrix}$
LIS_3	$\begin{bmatrix} 0.5567e^{i2\pi(0.1296)}, \\ 0.6364e^{i2\pi(0.0249)} \end{bmatrix}$	$\begin{bmatrix} 0.4125e^{i2\pi(0.5894)}, \\ 0.7741e^{i2\pi(0.8741)} \end{bmatrix}$
LIS_4	$\begin{bmatrix} 0.2363e^{i2\pi(0.0236)}, \\ 0.9678e^{i2\pi(0.0219)} \end{bmatrix}$	$\begin{bmatrix} 0.8123e^{i2\pi(0.8796)}, \\ 0.8254e^{i2\pi(0.9871)} \end{bmatrix}$

Output Layer

Step 1: We use the following formulas to determine the hidden layer criteria weights such that $\vartheta_j = (0.23,0.24,0.25,0.28)$.

Output layer

Step 2: Define the CFFSSTWA operator

Table 3, CFF SSTHWA operator

	HIC_1	HIC_2
LIS_1	$\begin{bmatrix} 0.1091e^{i2\pi(0.0245)}, \\ 0.3881e^{i2\pi(0.0212)} \end{bmatrix}$	$\begin{bmatrix} 0.3092e^{i2\pi(0.0456)}, \\ 0.5066e^{i2\pi(0.0269)} \end{bmatrix}$
LIS_2	$\begin{bmatrix} 0.1523e^{i2\pi(0.1698)}, \\ 0.1995e^{i2\pi(0.0298)} \end{bmatrix}$	$\begin{bmatrix} 0.0031e^{i2\pi(0.0172)}, \\ 0.8162e^{i2\pi(0.0662)} \end{bmatrix}$
LIS_3	$\begin{bmatrix} 0.1916e^{i2\pi(0.0217)}, \\ 0.2361e^{i2\pi(0.0126)} \end{bmatrix}$	$\begin{bmatrix} 0.6891e^{i2\pi(0.0963)}, \\ 0.7784e^{i2\pi(0.1297)} \end{bmatrix}$
LIS_4	$\begin{bmatrix} 0.0341e^{i2\pi(0.1234)}, \\ 0.0675e^{i2\pi(0.5431)} \end{bmatrix}$	$\begin{bmatrix} 0.3696e^{i2\pi(0.4596)}, \\ 0.5968e^{i2\pi(0.0369)} \end{bmatrix}$

	HIC_3	HIC_4
LIS_1	$\begin{bmatrix} 0.0198e^{i2\pi(0.0198)}, \\ 0.4911e^{i2\pi(0.0563)} \end{bmatrix}$	$\begin{bmatrix} 0.2991e^{i2\pi(0.0297)}, \\ 0.9913e^{i2\pi(0.1298)} \end{bmatrix}$
LIS_2	$\begin{bmatrix} 0.8031e^{i2\pi(0.0278)}, \\ 0.9142e^{i2\pi(0.0212)} \end{bmatrix}$	$\begin{bmatrix} 0.7402e^{i2\pi(0.7402)}, \\ 0.9741e^{i2\pi(0.9741)} \end{bmatrix}$
LIS_3	$\begin{bmatrix} 0.5567e^{i2\pi(0.1296)}, \\ 0.6364e^{i2\pi(0.0249)} \end{bmatrix}$	$\begin{bmatrix} 0.4125e^{i2\pi(0.5894)}, \\ 0.7741e^{i2\pi(0.8741)} \end{bmatrix}$
LIS_4	$\begin{bmatrix} 0.2363e^{i2\pi(0.0236)}, \\ 0.9678e^{i2\pi(0.0219)} \end{bmatrix}$	$\begin{bmatrix} 0.8123e^{i2\pi(0.8796)}, \\ 0.8254e^{i2\pi(0.9871)} \end{bmatrix}$

Step 3: The output layer generates score values by applying a score function.

CFFSSTWA operator Table 4.

CFFSSTWA operator Table 4

LIS_1	$[0.7844e^{i2\pi(0.1789)}, 0.3781e^{i2\pi(0.4115)}]$
LIS_2	$[0.5693e^{i2\pi(0.0459)}, 0.1785e^{i2\pi(0.7841)}]$
LIS_3	$[0.2245e^{i2\pi(0.2348)}, 0.2001e^{i2\pi(0.1987)}]$
LIS_4	$[0.2191e^{i2\pi(0.4471)}, 0.7819e^{i2\pi(0.0745)}]$

Step 4: Determine the ranking order.

$Z_1 = 0.0199,\ Z_2 = 0.5478,$

$Z_3 = 0.2979,\ Z_4 = 0.5148.$

5.2 Comparison technique with existing method

In this subsection, the comparison table is given below Table 5 (Fig. 3)

Different existing techniques Table 5

Methods	Score function	Ranking	Final Ranking
CFF [2]	$\begin{bmatrix} Z_1 = 0.2014, \\ Z_2 = 0.8755, \\ Z_3 = 0.5811, \\ Z_4 = 0.6712 \end{bmatrix}$	$\begin{bmatrix} Z_2 > \\ Z_4 > \\ Z_3 > \\ Z_1 \end{bmatrix}$	$\begin{bmatrix} Z_2 > \\ Z_4 > \\ Z_3 > \\ Z_1 \end{bmatrix}$
FFSW [15]	$\begin{bmatrix} Z_1 = 0.1789, \\ Z_2 = 0.6971, \\ Z_3 = 0.4668, \\ Z_4 = 0.6697 \end{bmatrix}$	$\begin{bmatrix} Z_2 > \\ Z_4 > \\ Z_3 > \\ Z_1 \end{bmatrix}$	$\begin{bmatrix} Z_2 > \\ Z_4 > \\ Z_3 > \\ Z_1 \end{bmatrix}$
FPF [12]	$\begin{bmatrix} Z_1 = 0.1821, \\ Z_2 = 0.9106, \\ Z_3 = 0.2019, \\ Z_4 = 0.6119 \end{bmatrix}$	$\begin{bmatrix} Z_2 > \\ Z_4 > \\ Z_3 > \\ Z_1 \end{bmatrix}$	$\begin{bmatrix} Z_2 > \\ Z_4 > \\ Z_3 > \\ Z_1 \end{bmatrix}$
FFS [14]	$\begin{bmatrix} Z_1 = 0.2981, \\ Z_2 = 0.7606, \\ Z_3 = 0.3678, \\ Z_4 = 0.5562 \end{bmatrix}$	$\begin{bmatrix} Z_2 > \\ Z_4 > \\ Z_3 > \\ Z_1 \end{bmatrix}$	$\begin{bmatrix} Z_2 > \\ Z_4 > \\ Z_3 > \\ Z_1 \end{bmatrix}$

5.3 Advantage of complex fermatean fuzzy sets

An improved version of traditional fuzzy set models, the Complex Fermatean Fuzzy Set (CFFS) was created to better represent the complex ambiguity and reluctance found in real-world decision-making situations. It was chosen for this study because of the following main benefits: CFFS generalizes the Fermatean fuzzy set by enabling complex-valued membership and non-membership degrees, providing more flexibility in expressing uncertain and imprecise information. This comes in handy in situations where human judgment cannot be quantified exactly, like expert assessments in MADM.

In classical fuzzy models, the degree of hesitation is indirectly obtained and restricted. But CFFS can inherently support hesitant and conflicting opinions due to its expanded structure. This is perfectly suitable for decision-making situations where decision-makers can agree and disagree on an option partially at a time, or be doubtful.

The use of a complex domain (imaginary and real parts) facilitates modeling of periodic, cyclic, or phase-shifted preferences—e.g., in time-dependent choice, subjective inference, or context-sensitive opinions. The richer semantic interpretation for expert input comes with the complex representation. CFFS can accommodate more discriminative aggregation operators, thereby providing improved discrimination among alternatives in MADM problems.

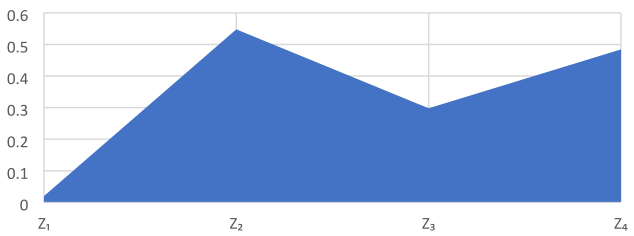


Fig. 3 Score function of different ranking

Numerous real-world decision-making issues are concerned with imprecise, vague, or incomplete information—particularly in finance, engineering, and human-oriented systems. CFFS offers more realistic modeling methods, allowing decision-makers to represent evaluations with intensity (magnitude) and direction. The fuzzy neural network model employed by the research in this work is enhanced by CFFS such that it embraces both quantitative and qualitative uncertainty during the learning process. The synergy improves the model's learning efficiency and adaptability in high-uncertainty situations.

5.4 Limitations

Notwithstanding the encouraging outcomes and sturdy design of the suggested decision-making framework, it is important to recognize several limitations:

1. The combination of several intricate elements, including fuzzy neural networks and CFFS aggregation operators, raises the computational requirements. The model may need a significant amount of memory and processing time for large-scale situations with numerous attributes and options.
2. The appropriate choice of parameters within the Schweizer–Sklar t-norm and t-conorm families determines how well the suggested aggregation operators operate. The quality and dependability of the results could be impacted by arbitrary or incorrect parameter selections.
3. Proper training data and hyperparameter tweaking are necessary for the Fuzzy Neural Network component. The model's capacity to learn may be hindered in fields with a lack of labeled fuzzy data or historical data.

5.5 Advantage of Schweizer–Sklar t-norm and t-conorm

The reason for the choice of Schweizer–Sklar t-norm and t-conorm in this research lies in their parametric versatility and generalized structure, which render them most appropriate to represent fuzzy interactions across a broad array of levels of uncertainty.

The Schweizer–Sklar norms generalize several common fuzzy operators. For example:

When the parameter $p \to 0$, the t-norm approximates the product t-norm.

When $p \to \infty$ it behaves like the minimum t-norm.

p enables the system designer or decision-maker to regulate the intersection (t-norm) and union (t-conorm) operations strictness. Such flexibility is of great utility in contexts such as Complex Fermatean Fuzzy Sets, where the strength of interaction between sources of information can be variable.

These norms are continuous, associative, and monotonic, which are important for mathematical coherence in fuzzy aggregation and in meeting the needs of fuzzy logic systems.

Human choice tends to lie between strict logic (min/max) and complete probabilistic interaction (product/sum). Schweizer–Sklar operators fill this gap, making it possible to better model expert information under sophisticated and imprecise circumstances.

5.6 Validity

The following crucial elements comprise the suggested CFFNN approach's strength:

It models intricate interdependencies across criteria that conventional linear models could miss by integrating nonlinear aggregation through a fuzzy neural structure. A generalized and flexible fusion method that can be adjusted to different data patterns is ensured by the use of Schweizer–Sklar t-norms and t-conorms in the hidden and output layers.

Because the Complex Fermatean fuzzy sets adequately capture uncertainty and ambiguity, the model is especially well-suited to decision-making scenarios containing these elements.

5.7 Sensitivity analysis

In this subsection, we define the sensitivity analysis

parameters	Methods	Ranking
q = 0.1	$Z_1 = 0.3814$, $Z_2 = 0.8155$, $Z_3 = 0.5991$, $Z_4 = 0.7719$	$[Z_2 > Z_4 > Z_3 > Z_1]$
0.2	$Z_1 = 0.0319$, $Z_2 = 0.9015$, $Z_3 = 0.6371$, $Z_4 = 0.7998$	$[Z_2 > Z_4 > Z_3 > Z_1]$
0.3	$Z_1 = 0.3008$, $Z_2 = 0.8201$, $Z_3 = 0.5103$, $Z_4 = 0.6908$	$[Z_2 > Z_4 > Z_3 > Z_1]$
0.4	$Z_1 = 0.0004$, $Z_2 = 0.6015$, $Z_3 = 0.4503$, $Z_4 = 0.5698$	$[Z_2 > Z_4 > Z_3 > Z_1]$
0.5	$Z_1 = 0.0023$, $Z_2 = 0.7896$, $Z_3 = 0.5509$, $Z_4 = 0.6872$	$[Z_2 > Z_4 > Z_3 > Z_1]$
0.6	$Z_1 = 0.1117$, $Z_2 = 0.4905$, $Z_3 = 0.3508$, $Z_4 = 0.4627$	$[Z_2 > Z_4 > Z_3 > Z_1]$

6 Conclusion

This research introduces a strong multi-attribute decision-making (MADM) system by merging Fuzzy Neural Networks (FNN) with Complex Fermatean Fuzzy Sets (CFFS). The system leverages the sophisticated uncertainty, indeterminacy, and hesitancy management capabilities of CFFS, linking these with FNN's learning and adaptability features to develop an intelligent and dynamic decision support framework. Three aggregation operators, including CFFSSTWA, CFFSSTOWA, and CFFSSTHWA, have been proposed on the basis of Schweizer–Sklar t-norms and t-conorms, which increases the flexibility of information fusion in complicated fuzzy situations. The operators are of significant importance for expressing decision-makers' behavior and preferences. The system is further enriched with the CFF-AHP approach, which is specifically tailored for procedure selection for remarkable innovativeness. A comprehensive case study and numerical example were performed to ensure the practicability of the model. Comparative analysis with the current approaches indicates that the proposed method has improved accuracy, greater flexibility, and better consistency in decision outcomes. The addition of the Fuzzy Neural Network adds a learning capability to the system that enables it to dynamically process fuzzy data and learn as additional data becomes available. This renders the suggested model optimally suited for real-time, large-scale, and intricate decision-making situations.

Future work, we intend to investigate the incorporation of other aggregation techniques, such as the Power Average and Muirhead Mean operators, into the CFFS–FNN framework in subsequent work. The model's adaptability and practical usefulness are further supported by its prospective applicability to other complex domains, including risk analysis, investment evaluation, and supplier selection.

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Data availability There is no associated data with this research. All data utilized in this study are original and provided within the article.

Declaration

Conflict of interest The authors declare that they have no known competing financial Interests or personal relationships that could have appeared to influence the work reported in this paper.

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